

Burraq Ur Rehman

Full-Stack Developer, Ruby on Rails, Python and AI Systems

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PROFESSIONAL SUMMARY

Full-stack engineer with 5+ years shipping production Ruby on Rails, Python/FastAPI, React, and Node.js applications for international product teams. Builds multi-tenant SaaS platforms and production AI features, including real-time voice agents, LLM tool calling, transcript pipelines, and vector search, behind provider-agnostic abstractions that survive vendor changes. Owns features end to end from data model to deployed interface, mentors junior engineers, and drives CI/CD and code-review standards that shorten release cycles.

TECHNICAL SKILLS

Languages: Ruby, Python, TypeScript, JavaScript, Node.js, SQL, HTML5, CSS3

Frameworks: Ruby on Rails, FastAPI, NestJS, Next.js, Node.js, Express.js, Django, Flask, Celery, Pydantic v2, SQLAlchemy

Frontend: React 18 and 19, Angular, Vite, TanStack Query, Redux, Context API, Tailwind CSS, shadcn/ui, Bootstrap, jQuery, responsive and accessible UI

Backend: RESTful API design, microservices, multi-tenant architecture, WebSockets, Socket.IO, background jobs, RBAC, JWT and JWKS authentication (Auth0, Devise, Passport), Stripe billing, third-party integrations

Databases: PostgreSQL, pgvector, SQLAlchemy 2 (async), Prisma, Alembic, Redis, NoSQL, query and schema optimization

Cloud: AWS (EC2, S3, Lambda, RDS, CloudWatch alarming), Vercel, Railway

DevOps: Docker, CI/CD pipelines, GitHub Actions, Nginx, PM2, Git, GitHub, GitLab, Bitbucket

Testing and Quality: pytest, RSpec, Playwright, Vitest, automated testing, ruff, mypy, ESLint, RuboCop, debugging, code review

AI and Machine Learning: LLM integration, function and tool calling, real-time voice AI agents (Retell, x.AI Grok realtime), speech pipelines over Twilio Media Streams, call transcript processing, vector search and embeddings (pgvector), retrieval-augmented generation, prompt design, provider-agnostic model abstractions, scikit-learn basics

PROFESSIONAL EXPERIENCE

Full-Stack Developer, Xprolabs

Lahore, Pakistan | Oct 2021 - Present

- Shipped and maintained multiple in-house modules end to end using TypeScript, React, Ruby on Rails, and Node.js, directly supporting product roadmap delivery.
- Built CI/CD pipelines with automated testing that cut deployment time by 40% and raised release quality.
- Optimized complex SQL queries and backend architecture, reducing application latency and improving response times by 35%.
- Designed and shipped RESTful APIs in Node.js powering React front ends across several product surfaces.
- Built internal tooling that streamlined engineering workflows and lifted developer productivity.
- Partnered with designers, product managers, and backend engineers to translate requirements into user-centric features.
- Mentored 3 junior developers on coding standards and code review, achieving 100% team retention.

Associate Software Engineer, Southville Solutions

Lahore, Pakistan | Mar 2021 - Oct 2021

- Delivered features across Node.js, JavaScript, TypeScript, React, and Angular within agile sprint cycles.
- Built responsive UIs with HTML5, CSS3, and Bootstrap, translating designs into production-ready components.
- Ramped quickly on new frameworks and tools, becoming a productive contributor within the first sprint cycles.

KEY ACHIEVEMENTS

- **CMMS Launch Contributor, Xprolabs (2023):** Led the launch of a computerized maintenance management system, improving operational efficiency by 45%.
- **Performance Excellence Award, Xprolabs (2023):** Recognized for optimizing application response times by 35%.
- **Team Leadership Recognition, Xprolabs (2023):** Mentored 3 developers with a 100% retention rate.
- **Programming Competition Winner, UOE Tech Fest (2020):** 1st place in a university-wide coding competition.

PROJECTS

ComboTutors, Multi-Tenant Online Tutoring Platform (LMS and CRM)

- Architected and shipped a multi-tenant SaaS platform running an online tutoring academy end to end, covering leads, scheduling, Google Meet classes, attendance, Stripe billing, and tutor payouts across four role classes, with the backend as the sole authorization boundary.
- Built a 373-endpoint backend with 176 migrations under an expand, migrate, contract policy, an idempotent billing engine issuing exactly one Stripe invoice per family per month, and pre-deploy quality gates of 303 pytest modules, 86 Vitest suites, and 20 Playwright flows. Authored roughly 995 of 1,656 commits across both repositories.

Stack: Next.js 16, React 19, TypeScript, Tailwind CSS v4, FastAPI, Python 3.14, SQLAlchemy, Pydantic v2, PostgreSQL, pgvector, Redis, Alembic, Stripe, Google Meet API, Docker, Vercel, Railway, GitHub Actions, Playwright, Vitest, pytest

Otto Revenue Engine, Multi-Tenant AI Voice and CRM SaaS for Automotive Dealerships

- Built a multi-tenant SaaS platform that automates inbound dealership calls with an AI voice agent, recovering revenue lost to missed calls and dropped follow-ups.
- Engineered the voice pipeline with signature-verified telephony webhooks, real-time tool calling for appointment booking, and call transcript persistence, across two providers (Retell and x.AI Grok realtime over Twilio Media Streams) behind a switchable abstraction for zero-downtime migration.

Stack: Python, FastAPI, async SQLAlchemy, PostgreSQL, Alembic, Redis, Celery, React 18, TypeScript, TanStack Query, Tailwind CSS, Auth0, Twilio, AWS S3, WebSockets, Docker

Pigeon Premier League, Tournament Management and Live Scoring Platform

- Built and deployed a production platform for endurance pigeon racing with live leaderboards, race-day time capture, and a permission-scoped admin panel, a full rewrite of a legacy system carrying 17 documented correctness and security defects.
- Designed a three-process architecture (API, pg-boss worker, Nginx-served SPA) on PostgreSQL alone with no Redis or external broker, bridging WebSocket JSON deltas across processes via LISTEN/NOTIFY, and covered it with 128 tests and a GitHub Actions pipeline that typechecks, migrates, tests, audits dependencies, and auto-deploys over SSH.

Stack: TypeScript, NestJS 11, Prisma 6, PostgreSQL, React 18, Vite 6, Tailwind CSS, TanStack Query v5, Socket.IO, pg-boss, JWT, Passport, Nginx, PM2, GitHub Actions

Kinari, Multilingual Survey Platform

- Built a dynamic survey builder in React and Node.js with multilingual creation, on-the-fly translation, and CSV import and export.
- Added result visualization, user impersonation, messaging, background jobs, and URL-based multi-step surveys.

Supply Chain Management System

- Architected a multi-tenant platform from scratch with Devise-based multi-profile logins for manufacturers and buyers.
- Delivered an e-commerce store, subscription plans, reviews, and manufacturer data scraping, deployed on AWS (S3, RDS) with TypeScript for scalability.

Mr. Cooper Apollo, Loan Management Platform

- Developed Apollo Web, a microservices-based platform streamlining home loan buying, selling, and refinancing.
- Shipped Cash-back rewards, a Pre-Approval Tool, a Digital Loan Tracker, and a Close-On-Time Guarantee.

Winad, School and Translation Management Platform

- Delivered client-driven features in Ruby on Rails and Haml with project-scoped access control for translated file downloads, full RSpec coverage, and RuboCop-enforced code quality.

Career Camp, E-Learning Platform

- Owned end-to-end delivery of an AWS-deployed e-learning platform with AJAX-driven dynamic event creation and an admin panel managing courses, students, tutors, and employers.

EDUCATION

B.S. Information Technology, University of Education (UOE), Lahore

Sep 2017 - Sep 2021

Relevant coursework: Software Engineering, Database Systems, Web Development, Data Structures and Algorithms, Networking.

CERTIFICATIONS

AWS Certified Developer, Associate. Udemy, May 2025

Learning SQL Programming. LinkedIn Learning, May 2024